Roll Your Own LAMBDA

How to describe the Excel function you want, so Claude Code builds and tests it.

You don't need to write LAMBDA syntax to get a LAMBDA you can trust. You need to **describe the function well**: what goes in, what comes out, how each number is defined, what counts as bad input, and what it must match. Claude Code, with the 5G skills installed, does the rest. It writes the formula, tests it in a real copy of Excel against an independent calculation, and won't call it finished until it ties out.

This article walks through one complete example. The function is `Example1`, a statistics table for Monte Carlo results, rebuilt from a description and checked cell by cell against a working function that already exists in the workbook. Then it's your turn.

1. The example workbook

`Roll Your Own LAMBDA Example.xlsx` holds one sheet, *LAMBDA Example*:

- **B3** spills 10,000 simulated project costs: `=fx.RiskPertλ(80, 100, 150, MC_Trials, 3, 8675309)`. The seed makes them reproducible.
- **C3** and **D3:D4** compare each trial with the budget in D2.
- **F3** is the *existing example LAMBDA*: `=fx.RiskStatsDetailλ(B3#, , B2)` spills a 33-row statistics table into F3:G35.
- **I3** is empty. That's where the new LAMBDA goes.

=fx.RiskPertλ(80, 100, 150, MC_Trials, 3, 8675309)		
Project	Budget check	120
121	Over	Trials at or under budget: 85.9%
110	Under	Trials over budget: 14.1%
99	Under	
90	Under	
95	Under	
112	Under	
114	Under	
91	Under	

Columns B to D: the simulated trials, the budget check and its two summary cells. The F3:G35 table appears in section 6, beside the new one.

Why rebuild a function that already exists?

Because it gives you a perfect answer key. Every cell Claude produces can be checked against F3:G35, so you learn what a good description looks like without having to trust anything. Once the method works, you point it at functions that don't exist yet (section 7).

2. Before you start

- Claude Code installed, with the **excel-dev** plugin from github.com/wfphillips128/claude-skills. Its docs/SETUP.md covers the rest: Python with `pip install -r requirements.txt`, and the Excel Labs add-in.
- Windows and Excel for Microsoft 365. The testing step drives a private copy of Excel in the background; your open workbooks aren't touched.
- The workbook in a folder of its own. Open a terminal in that folder and type `claude`. Everything below is typed into Claude Code, not into Excel.

3. What a good description covers

Most bad LAMBDAs come from a vague request, not bad syntax. "Give me summary statistics" leaves Claude guessing about a dozen things: which standard deviation, which percentile method, what order, what to do with blanks. Each guess is a chance to be plausibly wrong. Cover these six things and there's little left to guess:

Part	What to say	In the Example1 prompt
1. Name and place	The function's name and the exact call in its cell, with # for spilled ranges	`=Example1(B3#, , B2)` in I3
2. Inputs	Each argument: required or optional, its shape, its default	Trials, [Percentiles], [Name]
3. Output	Every row and column, in order, with labels	2 columns, 33 rows, listed
4. Definitions	Any number that could be computed more than one way	Sample vs population statistics; **the Mode rule**
5. Rules	How much documentation; self-contained; what is invalid and what to return	Plain LAMBDA; no calls to fx.*; #VALUE! below 4 numbers
6. Proof	What it must match, and how to show it	Match F3:G35; test in Excel; save a copy

Part 4 is the one people skip and the one that matters most. Here it's the **Mode**. Simulated values almost never repeat, so Excel's MODE.SNGL returns #N/A. The existing function bins the data and reports the midpoint of the tallest bin. Without that sentence, Claude would have to pick a method, and the Mode row would almost certainly not match.

4. The prompt

This is the whole instruction. Paste it into Claude Code as it stands:

PROMPT FOR CLAUDE CODE

```
Use the compose-5g-lambda skill.

Workbook: "Roll Your Own LAMBDA Example.xlsx" in this folder,
sheet "LAMBDA Example".

GOAL
Build a new LAMBDA named Example1 that produces the same table as the
existing fx.RiskStatsDetailλ in F3 (F3:G35), so I can check yours
against it cell by cell. It goes in cell I3 as:
    =Example1(B3#, , B2)

INPUTS
- Trials (required): the simulation results. A range or spilled
  array, any shape. Ignore blanks and text.
- Percentiles (optional): which percentiles to report, as fractions
  0 to 1. Default 5%, 10%, ... 95%.
- Name (optional): heading for the value column. Default "Value".

OUTPUT -- two columns, 33 rows with the defaults
- Row 1: blank | Name
- Then: Trials (count), Mean, Standard Error, Median, Mode, Standard
  Deviation, Sample Variance, Kurtosis, Skewness, Range, Minimum,
  Maximum, Sum. Sample statistics: STDEV.S, VAR.S, KURT, SKEW.
- Then one row per percentile, labelled P5, P10 ... (PERCENTILE.INC).

DEFINITIONS THAT MATTER
- Mode: simulated values rarely repeat. If at most half the values
  are distinct, use MODE.SNGL. Otherwise split the range into
  ROUND(SQRT(n)) equal-width bins (between 1 and 100) and return the
  midpoint of the tallest bin (the first one if two tie).

RULES
- Keep it simple. Full 5G documentation is overkill here: no
  documentation header block, no inline help table, no per-argument
  error messages. A plain LAMBDA with named LET steps and short
  comments.
- Self-contained: do not call fx.RiskStatsDetailλ, fx.RiskModeλ or
  any other function in the workbook.
- Return #VALUE! if Trials has fewer than 4 numbers, or if they are
  all the same (skewness and kurtosis can't be calculated).
- Before writing any formula, restate this spec and wait for my OK.

PROOF
- Test it in Excel against an independent calculation, including
  small and oddly shaped inputs, and show me that
  =Example1(B3#, , B2) matches F3:G35.
- Do not change my workbook. Save a copy named
  "Roll Your Own LAMBDA Example - Completed.xlsx" with the function
  installed at I3.
```

Four lines in it do more work than they look:

- **"Use the compose-5g-lambda skill."** This loads the build discipline: the testing harness, the list of known Excel traps, and the rule that nothing ships until it ties out. Without it you get a formula nobody has run.
- **"Keep it simple. Full 5G documentation is overkill here."** By default the skills build to Craig Hatmaker's full 5G standard. Say so when you want less (see the box below).
- **"Restate this spec and wait for my OK."** You catch misunderstandings while they're a sentence, not a formula.
- **"Do not change my workbook."** The skill works paste-only by default; asking for a named copy gives you a finished file to open without risking the original.

Full 5G documentation, or a plain LAMBDA?

Full 5G adds a documentation header, an inline help table (type the function with no arguments and it explains itself), and a named message for each bad argument. That is worth the extra length for a function you'll share or keep in a library.

For a one-workbook function like this one it's overkill: about twice the length, and harder to read while you're learning what the formula does. Ask for whichever fits, and say it in the prompt. Built the full way, this same function came to 2,332 characters; the plain version is 1,315.

5. What Claude Code does with it

These are the steps from the run that produced this article. Expect the same shape when you do it.

Step	What happens	What you see
Reads the workbook	Opens it read-only and lists every LAMBDA already defined, with its arguments. It notes that fx.RiskStatsDetailλ calls two helpers, which is why the prompt says self-contained.	A short inventory
Restates the spec	Arguments, output rows, the Mode rule, the #VALUE! rule, the benchmark.	A table to approve, or to correct
Writes the function	A plain LAMBDA: one LET with a named step per idea (inputs, statistics, Mode, percentiles, assembly), short comments, and a last step that is a name so any step can be inspected.	1,315 characters of formula
Checks its own yardstick	Writes an independent Python calculation from the statistical definitions, then confirms that calculation reproduces F3:G35 before using it to judge anything.	All 33 rows; worst gap 5.5e-13 (rounding noise)
Tests in Excel	Installs the function in a hidden, private Excel and compares it with the Python calculation over hand-picked and random cases: tiny inputs, 2-D ranges, repeated values, text mixed in, custom percentiles, bad input.	PASS: 121 of 121 checks
Proves it in your workbook	Saves the copy, enters the formula at I3, recalculates, and compares all 66 cells with F3:G35. Adds a tie-out cell at L3.	66 of 66 match; L3 says "Ties to F3:G35"
Hands it over	The readable source (for Excel Labs), install steps, and the test results.	Example1.txt and the completed workbook

What the testing found

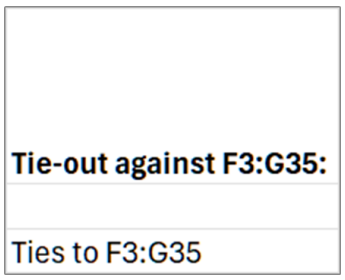
The first build stopped on a case the original description never mentioned: **every trial the same value**. Skewness and kurtosis can't be calculated then, so the table came back with errors in two rows.

The fix was a decision, not a formula: return #VALUE! for that case, as for fewer than four numbers. That is why the prompt above includes the rule. When this happens to you, Claude should bring it to you as a question. Add the answer to your description, so the next function handles it from the start. Finding these gaps is what the testing is for.

6. The result

Existing example LAMBDA:			Location for new target LAMBDA:	
	Project			Project
Trials	10,000		Trials	10,000
Mean	105.1		Mean	105.1
Standard Error	0.1		Standard Error	0.1
Median	103.9		Median	103.9
Mode	95.5		Mode	95.5
Standard Deviation	12.8		Standard Deviation	12.8
Sample Variance	164.7		Sample Variance	164.7
Kurtosis	(0.5169)		Kurtosis	(0.5169)
Skewness	0.3905		Skewness	0.3905
Range	67.2		Range	67.2
Minimum	80.4		Minimum	80.4
Maximum	147.6		Maximum	147.6
Sum	1,050,679.9		Sum	1,050,679.9
P5	86.4		P5	86.4
P10	88.9		P10	88.9
P15	91.2		P15	91.2
P20	93.1		P20	93.1
P25	94.9		P25	94.9
P30	96.6		P30	96.6
P35	98.5		P35	98.5
P40	100.2		P40	100.2
P45	102.0		P45	102.0
P50	103.9		P50	103.9
P55	105.7		P55	105.7
P60	107.7		P60	107.7
P65	109.7		P65	109.7
P70	111.9		P70	111.9
P75	114.1		P75	114.1
P80	116.5		P80	116.5
P85	119.4		P85	119.4
P90	123.1		P90	123.1
P95	128.0		P95	128.0

F3:G35 (the existing function) beside I3:J35 (Example1, built from the description).



The tie-out cell in L3 compares all 66 cells with F3:G35, within 1e-9. Both tables are spilled by a single formula.

Installing it yourself

- **Excel Labs (recommended):** open the Advanced Formula Environment, add a module, paste the contents of Example1.txt, and save. Comments and indentation are fine there; it strips them on the way in.

- **Name Manager:** Excel stores this function at 1,986 characters once it adds its own prefixes, and the Name Manager box won't accept a paste that long. Short functions can go there; for this one, use Excel Labs.
- Then type `=Example1(B3#, , B2)` in I3. The # means "the whole spill from B3", so the function picks up all 10,000 trials and follows the data if MC_Trials changes. The empty middle argument keeps the default percentiles.

7. Your turn

Copy the template, fill in the angle brackets, and keep the six parts. Two follow-on exercises use the same workbook, and each one teaches something new.

PROMPT TEMPLATE

```
Use the compose-5g-lambda skill.            (or convert-to-5g -- see below)

Workbook: "<file>.xlsx", sheet "<sheet>".

GOAL
Build a LAMBDA named <Name> that <does what, in one sentence>.
It goes in <cell> as: =<Name>(<arguments, using # for spills>)

INPUTS
- <Arg> (required/optional): <what it is, its shape>. Default <...>.

OUTPUT
- <rows and columns, in order, with labels>

DEFINITIONS THAT MATTER
- <any measure that could be computed two ways: say which>

RULES
- Documentation: <plain LAMBDA, no 5G documentation>  or
  <full 5G documentation: inline help and error messages>.
- Self-contained. <Anything it must not call or reference.>
- What counts as invalid input, and what to return.
- Before writing any formula, restate this spec and wait for my OK.

PROOF
- Benchmark: <cells / a grid / a hand calculation it must match>.
- Test it in Excel against an independent calculation, including
  edge cases (smallest input, odd shapes, optional arguments omitted).
- Do not change my workbook. Save a copy named "<...> - Completed.xlsx".
```

Exercise 2: replace a grid with a function (convert-to-5g)

Columns C and D are classic grid formulas: one row per trial, then two summary cells. Turning them into one function shows the **convert-to-5g** skill, which starts from formulas you already have rather than from a blank page. Its benchmark is the two percentages already on the sheet.

STARTER

```
Use the convert-to-5g skill.

In "Roll Your Own LAMBDA Example.xlsx", column C (from C3) marks each
trial "Under" or "Over" the budget in D2, and D3:D4 turn that into two
percentages. Replace all three with one LAMBDA named Example2, placed
in I38 as =Example2(B3#, D2), that spills a small table: Budget,
% at or under budget, % over budget, average overrun when over, worst
case, and the budget needed for 80% and 90% confidence. The two
percentages must match D3:D4 exactly. Plain LAMBDA, no 5G
documentation. <...rules and proof as in Example1...>
```

Exercise 3: something new, with a built-in check

No existing answer key this time, so the description has to supply its own proof: the counts must add up to the number of trials, and the cumulative % must end at 100%. Writing those checks into the prompt is how you make a new function testable.

STARTER

```
Use the compose-5g-lambda skill.

Build Example3(Trials, [Bins]) that returns a frequency table for a
histogram: Bin start, Bin end, Count, % of trials, cumulative %.
Default bins: ROUND(SQRT(n)) between 1 and 100, equal width, the last
bin closed on the right. Counts must add up to the number of trials
and the cumulative % must end at 100%. <...>
```

When something goes wrong

- **The skill didn't load** and you got an untested formula: name it ("use the compose-5g-lambda skill") and check `/plugin` lists excel-dev as enabled.
- **You got a help table and error messages you didn't want**: the skills default to full 5G. Add the "keep it simple" rule.
- **A number is close but not equal**: there's a definition you didn't state. Ask Claude which method it used for that row, decide, and add the sentence to your description.
- **#CALC! on small inputs**: a known Excel trap with empty arrays. The skill's pitfalls reference covers it, and asking for edge-case tests catches it.
- **It says it works but showed no test table**: ask for the test results. A function that hasn't been run against a benchmark isn't finished.

The 5G methodology, its templates and conventions are Craig Hatmaker's. See *Beyond Excel* (sites.google.com/site/beyondexcel) for the full treatment. The skills are an implementation of it.

Files that go with this article

File	What it is
Roll Your Own LAMBDA Example.xlsx	The starting template (unchanged)
Roll Your Own LAMBDA Example - Completed.xlsx	Example1 installed and working at I3, tie-out at L3
Roll Your Own LAMBDA\Example1.txt	The function, ready for Excel Labs
Roll Your Own LAMBDA\example1_harness.py	The test that ran it in Excel against the Python calculation
Roll Your Own LAMBDA\harness-run.txt	That test's output: 121 checks passed

Appendix: the finished function

Example1.txt, exactly as Claude delivered it: the readable form for Excel Labs, with comments and indentation.

```
// Example1: summary statistics and percentiles for a set of simulation trials.
// Built from a plain-English description; see "Roll Your Own LAMBDA.pdf".
//   =Example1(B3#, , B2)
Example1 = LAMBDA(Trials, [Percentiles], [Name],
    LET(
        // ---- Inputs -------------------------------------------------------
        Cells,     TOCOL(Trials, 3),                       // any shape -> one column; skip blanks and errors
        Flat,      FILTER(Cells, ISNUMBER(Cells), ""),      // numbers only (text is ignored, as AVERAGE does)
        Pcts,      IF(ISOMITTED(Percentiles), SEQUENCE(19, 1, 5, 5) / 100, TOCOL(Percentiles, 1)),
        Label,     IF(ISOMITTED(Name), "Value", INDEX(Name, 1, 1) & ""),

        // ---- Statistics ---------------------------------------------------
        nTrials,   ROWS(Flat),
        StdDev,    STDEV.S(Flat),

        // Mode. Simulated values rarely repeat, so MODE.SNGL alone returns #N/A.
        // If at most half the values are distinct, use MODE.SNGL. Otherwise bin
        // the data into ROUND(SQRT(n)) equal-width bins (1 to 100) and take the
        // midpoint of the tallest bin (the first one if two tie).
        Discrete,  ROWS(UNIQUE(Flat)) <= nTrials / 2,
        Bins,      MAX(1, MIN(100, ROUND(SQRT(nTrials), 0))),
        Lo,        MIN(Flat),
        Width,     (MAX(Flat) - Lo) / Bins,
        BinNo,     IF(Width > 0,
                      IF(Flat >= Lo + Bins * Width, Bins, INT((Flat - Lo) / Width) + 1),
                      1),
        Counts,    DROP(FREQUENCY(BinNo, SEQUENCE(Bins)), -1),
        ModeVal,   IF(Discrete, MODE.SNGL(Flat),
                      IF(Width > 0, Lo + (XMATCH(MAX(Counts), Counts) - 0.5) * Width, Lo)),

        PctValues, MAP(Pcts, LAMBDA(pct, PERCENTILE.INC(Flat, pct))),
        PctLabels, MAP(Pcts, LAMBDA(pct, "P" & ROUND(pct * 100, 4))),

        // ---- Assemble: labels beside values -------------------------------
        Labels,    VSTACK("",
                     {"Trials";"Mean";"Standard Error";"Median";"Mode";"Standard Deviation";
                      "Sample Variance";"Kurtosis";"Skewness";"Range";"Minimum";"Maximum";"Sum"},
                     PctLabels),
        Values,    VSTACK(Label, nTrials, AVERAGE(Flat), StdDev / SQRT(nTrials), MEDIAN(Flat),
                     ModeVal, StdDev, VAR.S(Flat), KURT(Flat), SKEW(Flat),
                     MAX(Flat) - MIN(Flat), MIN(Flat), MAX(Flat), SUM(Flat), PctValues),
        Result,    HSTACK(Labels, Values),

        // Kurtosis and skewness need at least 4 numbers that are not all the same.
        Output,    IF(OR(nTrials < 4, MAX(Flat) = MIN(Flat)), #VALUE!, Result),
        Output
    )
);
```

Appendix: a Name Manager version

The same function packed onto one line for a quick paste into Name Manager: no comments, short step names, 783 characters. It returns exactly the same table: it passed the same 121 checks in Excel and matches F3:G35 in all 66 cells.

NAME MANAGER FORMULA (Refers to)

```
=LAMBDA(Trials,[Percentiles],[Name],LET(v,TOCOL(Trials,3),x,FILTER(v,ISNUMBER(v),""),n,ROWS(x),p,IF(ISOMITTED(P
ercentiles),SEQUENCE(19,1,5,5)/100,TOCOL(Percentiles,1)),sd,STDEV.S(x),lo,MIN(x),hi,MAX(x),k,MAX(1,MIN(100,ROUN
D(SQRT(n),0))),w,(hi-lo)/k,b,IF(w>0,IF(x>=lo+k*w,k,INT((x-lo)/w)+1),1),f,DROP(FREQUENCY(b,SEQUENCE(k)),-1),md,I
F(ROWS(UNIQUE(x))<=n/2,MODE.SNGL(x),IF(w>0,lo+(XMATCH(MAX(f),f)-0.5)*w,lo)),t,HSTACK(VSTACK("",{"Trials";"Mean";
"Standard Error";"Median";"Mode";"Standard Deviation";"Sample
Variance";"Kurtosis";"Skewness";"Range";"Minimum";"Maximum";"Sum"},"P"&ROUND(p*100,4)),VSTACK(IF(ISOMITTED(Name)
,"Value",INDEX(Name,1,1)&""),n,AVERAGE(x),sd/SQRT(n),MEDIAN(x),md,sd,VAR.S(x),KURT(x),SKEW(x),hi-lo,lo,hi,SUM(x)
,PERCENTILE.INC(x,p))),IF(OR(n<4,hi=lo),#VALUE!,t)))
```

Copy it from the file, not from this page

A PDF adds line breaks where the formula wraps above, and a line break pasted into Name Manager can cut the formula short. Use the exact copy in **Example1-NameManager.txt**. It is also attached to this PDF: open the attachments (paperclip) pane in your PDF reader.

Install it

- **Formulas > Name Manager > New**.
- **Name:** Example1. **Comment** (optional): Summary statistics and percentiles for a set of simulation trials.
- **Refers to:** select everything in the box and paste the formula. It starts with `=LAMBDA(`.
- **OK**, then **Close**. In I3 type `=Example1(B3#, , B2)`.
- If Name Manager won't take the paste, use Excel Labs (section 6). Or skip naming it altogether: in I3, paste the formula and add `(B3#, , B2)` right after the final bracket, which calls the LAMBDA on the spot.

Reading the short names

Step	Means	Step	Means
v	every cell of Trials, as one column	k	number of bins
x	the numbers only	w	bin width
n	how many numbers	b	which bin each value falls in
p	the percentiles to report	f	how many values in each bin
sd	sample standard deviation	md	the Mode
lo, hi	minimum and maximum	t	the finished table

Two shortcuts make it shorter than the readable version without changing a result: the percentile labels and values are calculated on the whole list at once instead of through MAP, and each intermediate step has a one- or two-letter name. The letters r and c are avoided: Excel reserves them as names (they mean row and column).